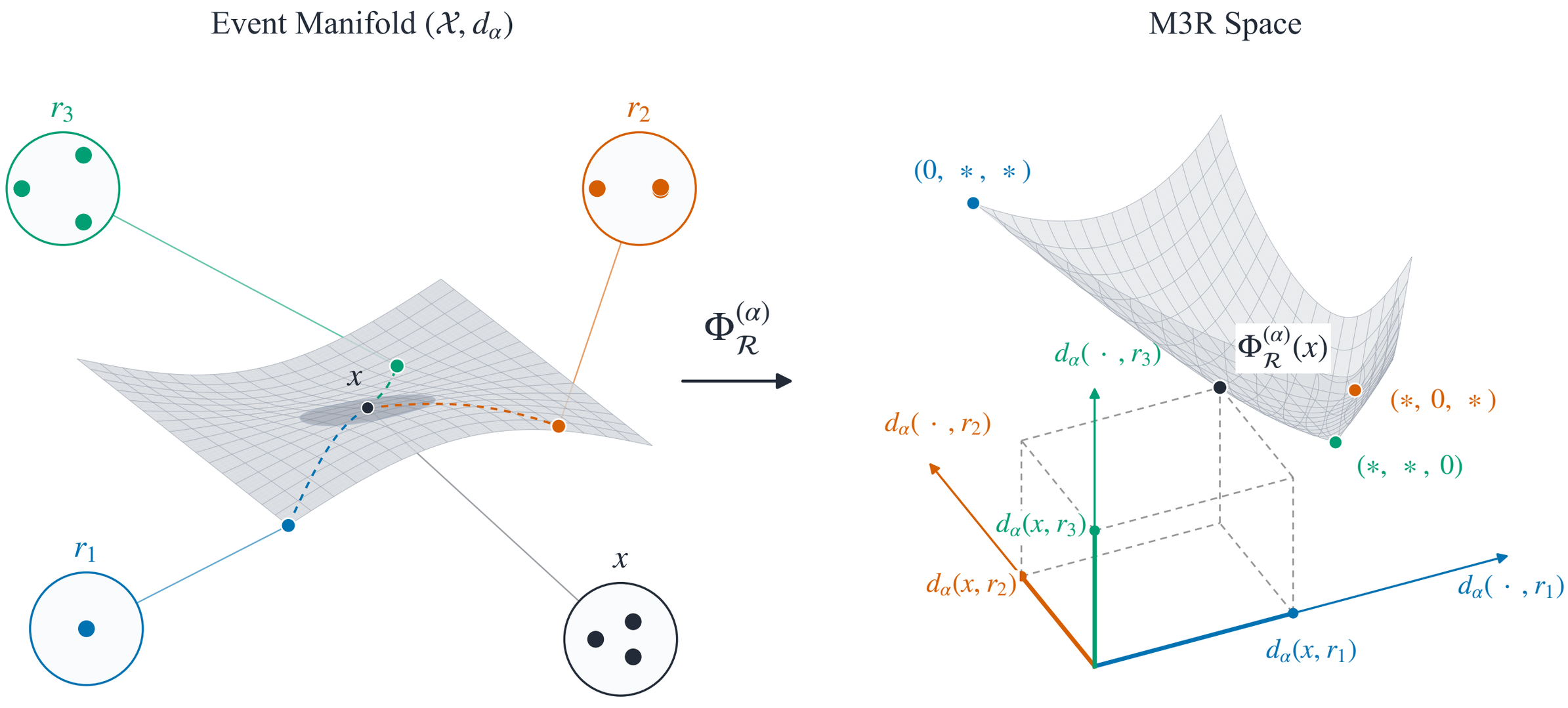

Event Manifold $(\mathcal{X}, d_\alpha)$
M3R Space
r_3
r_2
r_1
x
x
$\Phi_{\mathcal{R}}^{(\alpha)}$
$(0, *, *)$
$(*, 0, *)$
$(*, *, 0)$
$\Phi_{\mathcal{R}}^{(\alpha)}(x)$
$d_\alpha(\cdot, r_3)$
$d_\alpha(\cdot, r_2)$
$d_\alpha(\cdot, r_1)$
$d_\alpha(x, r_3)$
$d_\alpha(x, r_2)$
$d_\alpha(x, r_1)$
$x \mapsto (d_\alpha(x, r_1),\ d_\alpha(x, r_2),\ d_\alpha(x, r_3))$